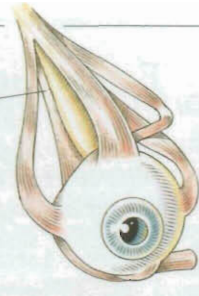
EYES AND SEEING

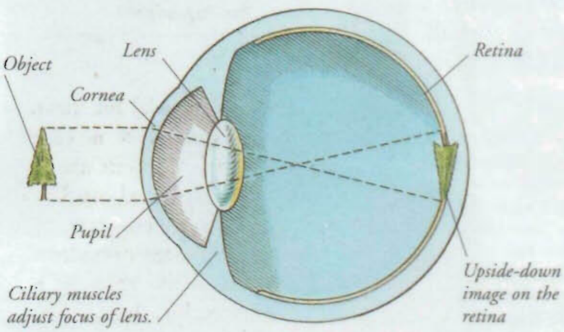
YOUR EYES ENABLE you to see by stimulating the creation of images in your brain. Each of the two eyeballs is a sphere measuring 6.25 cm (2.5 inches) in diameter. Eyeballs contain sensory cells that, when stimulated by light, send messages to the brain that are interpreted as images. Reflex mechanisms control the amount of light entering the eye, and enable the eye to focus on objects whether they are near or distant. Much of the eyeball is hidden because it is protected within the orbit, a bony socket in the skull. The delicate outer surface of the eye is also protected by the eyelids.

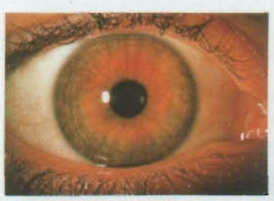


Moving the eye
Your eyes move constantly, even when you are staring. Six muscles move each eyeball and hold it in place inside its skull socket. Each muscle pulls the eye in a different direction, enabling the eyeball to move up and down, from side to side, and diagonally. Your brain controls these movements to make sure that both eyes move together.

Seeing

The eyes gather light from whatever you look at. The cornea and lens focus this light on the retina to produce an upside-down image. Cells inside the retina, called rods and cones, respond to light by sending nerve impulses along the optic nerve to the brain. The brain interprets these impulses so you see the image the right way up.





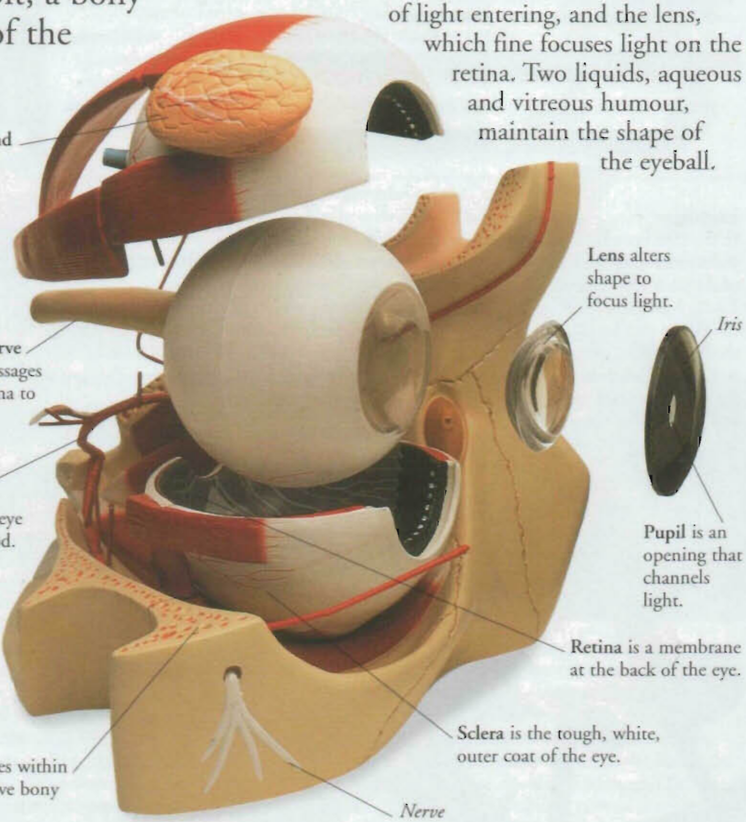
Iris and pupil
The iris controls the amount of light entering the eye. Muscles in the iris alter the size of the pupil, the opening that allows in light. In dim light, the pupil widens; in bright light, the pupil gets smaller.

Rods and cones
There are about 120 million rods and 7 million cones in the retina. Rods work best in dim light. Cones are responsible for colour vision and enable you to see things in detail.

Tears
Tears are released by lacrimal (tear) glands above the eye. When you blink, tears spread over the eye's surface. This keeps the cornea moist, washes away dust, and kills germs. After flowing over the eye, tears drain through two small openings at the side of the eye into the lacrimal duct, and then into the nose.

Inside the eye

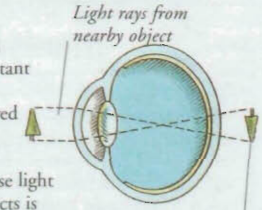
The transparent cornea at the front of the eye helps to focus light as it enters. Behind the cornea are the iris, which controls the amount of light entering, and the lens, which fine focuses light on the retina. Two liquids, aqueous and vitreous humour, maintain the shape of the eyeball.

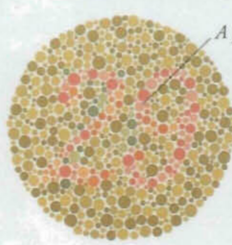


Vision defects

The most common vision problems are long sight and short sight. In both cases the eye does not focus light properly on the retina. Some people, mainly males, have colour blindness and cannot distinguish between certain colours, most often red and green.

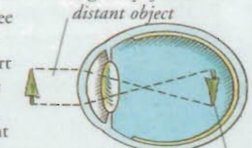
Long sight
A long-sighted person sees distant objects clearly, but has a blurred view of close objects. This happens because light from near objects is focused behind the retina, rather than on it.





A person with red-green colour blindness cannot see this number.

Short sight
A short-sighted person can see close objects clearly, but sees distant objects as a blur. Short sight is usually caused by the eyeball being longer than normal. This means that light entering the eye from distant objects is focused in front of the retina, rather than on it.



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